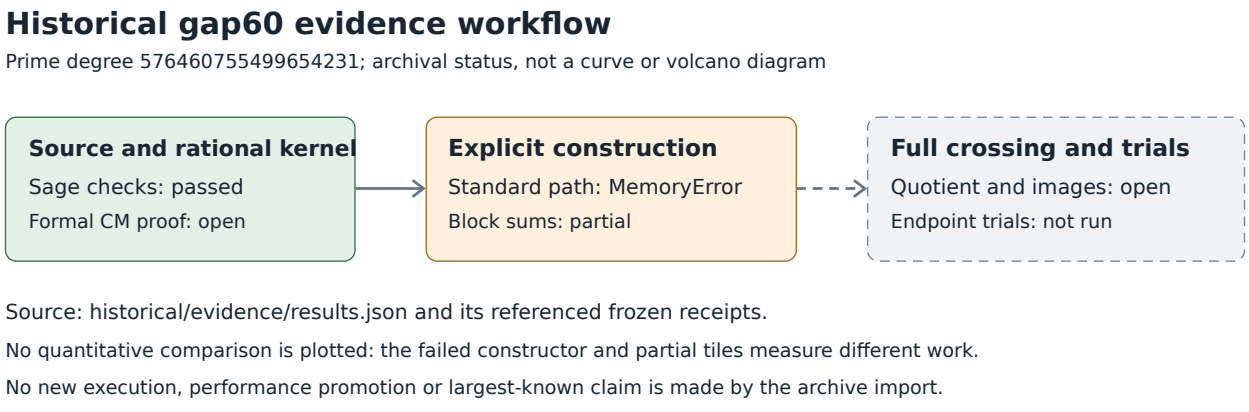
Historical 60-bit prime conductor-gap archive

This import preserves a verified source-and-kernel component, a failed explicit constructor, and checked partial polynomial sums. It adds no new experiment or native execution path. The frozen producer revision is `b6666f5058260f602070c06d1f56d03979295dec`.



What the historical records support

The archived prime degree is 576460755499654231 (60 bits). Exact field, source-model coefficients, subgroup and kernel coordinates are in `../parameters.json`; the unchanged summary is `../historical-summary.json`. The historical Sage replay passed its source, field, kernel-order and conductor-argument checks. These are executable checks and a CM/volcano argument, not a complete formal Lean proof.

The standard constructor failed with MemoryError. The alternative completed bounded pieces of the polynomial sums and matched complete small examples; its partial pieces are not a full map evaluation. The explicit quotient, complete non-kernel point images and matched endpoint ECDLP trials remain open. No quantitative comparison is plotted because the failed constructor and the partial tiles measure different work.

Provenance and retrieval

`../source-and-evidence.tar.xz.sha256` pins the immutable archive. Retrieve it with the repository artifact tool if migrated, verify with `sha256sum -c`, then extract it. The extracted `historical/evidence/input_manifest.json` checks original paths and bytes. Original Python/Sage source, native C helper, clocks, failures and receipts are historical evidence; this import does not install a Python research harness in crypto.

The source archive SHA-256 is: `e23bac98d4d3c663597fccaed07d19402f93ebeb0809f916b97b58309f6cbabd`.

Related work and remaining obligation

The existing problem record is `problems/conductor-gap-upper-floor-and-cost.md`. The archive's `historical/research/route_assessment.json` distinguishes initial map-image requirements, missing endpoint inputs and the kernel's trivial Frobenius extension orbit. It links the primary sources:

- Robert: Evaluating isogenies in polylogarithmic time.
- Galbraith: Climbing and descending tall volcanos.
- Banegas et al.: Fast and Frobenius.

The next technical obligation is an explicit quotient and non-kernel images for the unchanged minimum 60-bit target, with a feasible native execution path and fresh receipts. Archive publication does not discharge that obligation. No largest-known status, backend ranking or leaderboard row is promoted. Existing canonical graphs remain unchanged.